

HARBEL, Liberia

WMO: 656600

Lat: 6.234N

Lon: 10.362W

Elev: 9

StdP: 101.21

Time Zone: 0.00 (E00)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.0	21.9	17.9	12.9	24.6	19.8	14.5	25.0	7.6	26.6	7.3	26.8	1.3	40	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	8.8	33.2	26.8	33.0	26.8	32.2	26.6	28.1	32.0	27.7	31.5	27.4	31.1	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	30.9	26.8	22.5	30.6	26.2	21.7	29.7	90.2	32.0	88.4	31.9	87.1	31.4	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.3	5.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	27.5	28.1	28.2	28.0	27.6	26.4	25.6	25.3	25.9	26.9	27.4	27.5
	DBStd	1.31	1.06	0.96	0.80	0.85	0.90	1.01	0.85	0.82	0.89	0.88	0.71	0.83
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6215	543	508	565	540	544	492	483	475	478	522	522	542
	CDD18.3	3173	285	274	307	290	286	242	225	217	228	264	272	284
	CDH23.3	26700	2494	2583	3053	2895	2712	1857	1386	1199	1491	2070	2431	2529
	CDH26.7	8449	1012	1030	1243	1102	902	382	171	117	246	541	779	926
(14) Wind	WSAvg	1.9	1.6	1.8	2.0	2.0	1.9	1.9	2.1	2.1	1.9	1.9	1.6	1.5
(15) Precipitation	PrecAvg	2293	27	50	70	128	227	368	254	292	361	300	167	50
	PrecMax	2944	150	94	111	221	336	533	512	473	555	449	274	110
	PrecMin	1528	4	24	35	52	106	191	135	99	201	152	93	10
	PrecStd	280	27	17	18	44	61	81	95	89	89	70	46	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.2	34.1	33.2	33.0	31.1	30.0	29.8	30.9	32.0	32.8	33.2
		MCWB	25.4	26.7	27.0	27.7	27.7	26.9	26.0	25.9	26.9	27.0	27.4	26.5
	2%	DB	33.0	33.2	33.2	32.9	32.1	30.2	29.0	28.9	29.9	31.1	31.9	32.2
		MCWB	26.1	26.7	27.0	27.5	27.1	26.4	25.3	25.2	26.1	26.8	27.0	26.3
	5%	DB	32.2	32.9	32.9	32.1	31.7	29.9	28.2	28.0	29.0	30.2	31.1	31.9
		MCWB	25.9	26.6	27.0	27.1	27.0	26.2	25.0	24.9	25.7	26.3	26.8	26.2
	10%	DB	31.8	32.0	32.0	31.8	30.9	29.0	27.8	27.1	28.1	29.8	30.2	31.1
		MCWB	25.9	26.1	26.6	26.9	26.9	26.0	24.9	24.7	25.5	26.1	26.3	26.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.3	28.5	28.9	28.2	27.2	26.5	26.4	27.0	27.7	28.0	28.1
		MCDB	32.3	33.0	32.8	32.6	31.3	29.7	28.7	28.4	29.5	30.9	31.7	31.8
	2%	WB	27.3	27.6	27.9	28.0	27.6	26.8	26.0	25.8	26.5	27.0	27.2	27.4
		MCDB	31.4	32.1	32.1	31.6	30.9	29.4	28.3	27.8	28.9	30.2	30.8	31.1
	5%	WB	26.7	27.0	27.4	27.6	27.2	26.5	25.6	25.4	26.1	26.6	26.8	27.0
		MCDB	30.7	31.1	31.5	31.2	30.4	29.0	27.8	27.3	28.4	29.6	30.2	30.6
	10%	WB	26.4	26.5	26.8	27.0	26.8	26.1	25.2	25.0	25.7	26.3	26.6	26.6
		MCDB	30.1	30.4	30.7	30.7	29.9	28.5	27.2	26.8	27.8	29.1	29.8	29.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	8.8	8.8	8.1	7.1	5.5	4.5	4.1	5.1	6.5	7.2	8.3
		MCDBR	10.0	9.6	9.4	8.6	7.9	6.6	5.5	5.3	6.2	7.4	7.8	9.1
	5% WB	MCWBR	4.5	4.1	4.2	4.1	3.9	3.4	3.0	2.9	3.4	3.9	4.0	4.2
		MCWBR	9.1	9.0	8.8	8.3	7.4	6.0	5.1	4.7	5.7	7.1	7.5	8.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.738	0.723	0.737	0.599	0.521	0.513	0.496	0.481	0.466	0.487	0.551	0.654
		taud	1.618	1.673	1.656	1.946	2.148	2.174	2.230	2.282	2.342	2.269	2.049	1.788
	Ebn at Noon	Ebn at Noon	633	661	657	742	784	779	798	825	850	834	771	684
		Edh at Noon	265	256	262	193	154	147	140	136	130	139	171	220
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.97	5.05	5.18	5.09	4.73	3.96	3.86	3.82	4.10	4.64	4.64	4.68
	RadStd	RadStd	0.32	0.29	0.32	0.27	0.20	0.23	0.21	0.24	0.19	0.19	0.21	0.23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page